higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

CONTROL SYSTEMS N6

15 APRIL 2019

This marking guideline consists of 10 pages.

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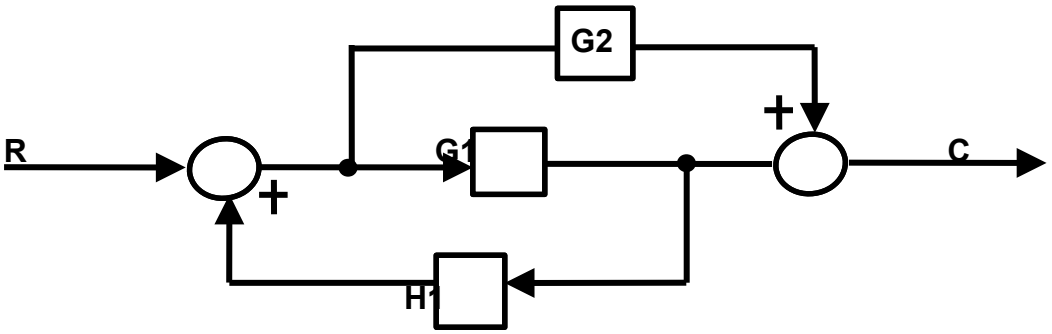
QUESTION 1

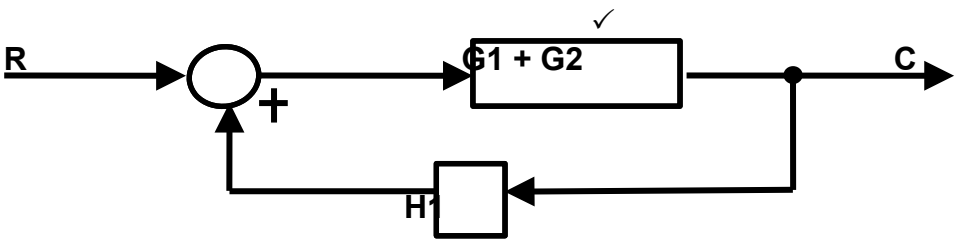
1.1 True
1.2 True
1.3 False
1.4 False
1.5 True
1.6 False
1.7 True
1.8 False
1.9 False
1.10 True

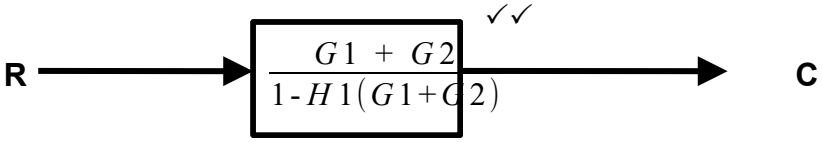
(10 × 1) **[10]**

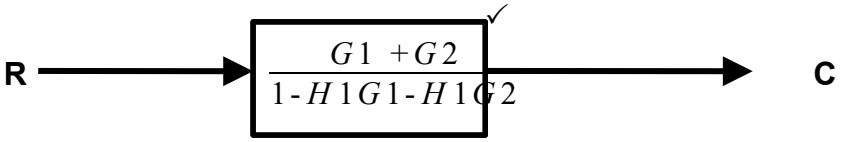
QUESTION 2

2.1









$$\frac{C}{R} = \frac{G1 + G2}{(1 - H1G1 - H1G2)}$$ ✓

(5)

2.2
- G – direct transfer function or forward transfer function
- H – feedback transfer function (2)

2.3
- They are used as an alternative method for calculating the transfer function of a control system.
- They are a shorthand pictorial representation of the cause and effect relationship between the input and the output of a physical system. (2)

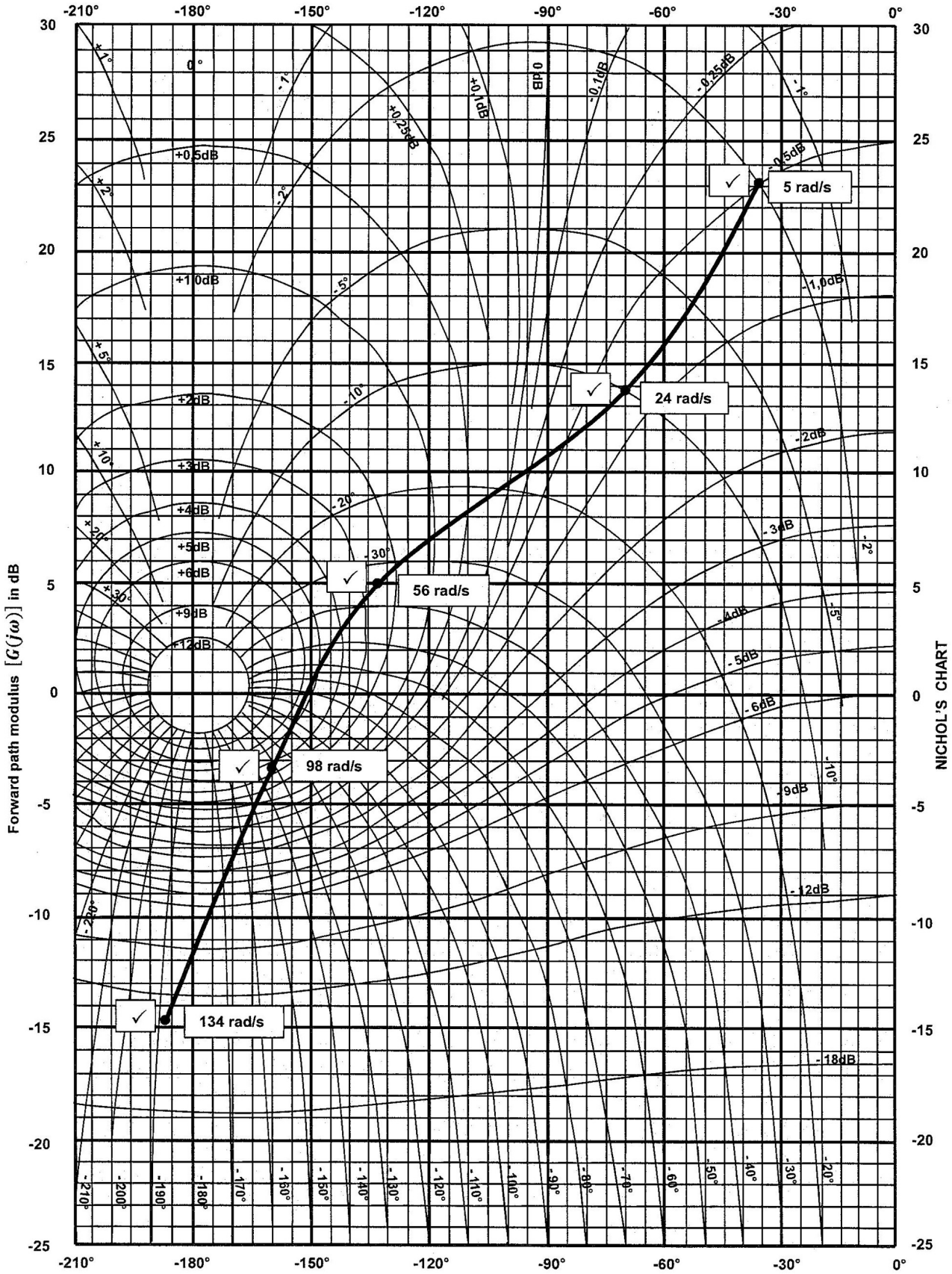
2.4 It allows for more than one signal to be fed into an amplifier at the same time. (1)
[10]

QUESTION 3

3.1 -44 dB/decade (1)
3.2 -66°/decade (1)
3.3 4 rad/s (1)
3.4 14 rad/s (1)
3.5 -113° (1)
3.6 -25 dB (1)
3.7 25 dB (1)
3.8 67° (1)
3.9 The gain and phase margins are positive, thus ensuring stability in the system. (2)
[10]

QUESTION 4

4.1



(1 mark for the plot of each gain/phase point) (5)

4.2

ω (rad/sec)	5	24	56	98	134
Magnitude in dB	23,2	13,8	5	-3,4	-14,7
Phase in degrees	-37°	-70°	-135°	-160°	-187°
	✓	✓	✓	✓	✓

(5)
[10]

QUESTION 5

$$G(s)H(s) = \frac{30\ AB}{S(S+30)}$$

5.1 Open-loop poles = 0 ; -30 ✓ Zeros = None✓ (2)

5.2
$$S_c = \frac{\sum Poles - \sum Zeros}{N_p - N_z}$$

$$= \frac{(0-30) - 0}{2 - 0} \quad \checkmark$$

$$= \frac{-30}{2}$$

$$S_c = -15 \checkmark$$ (2)

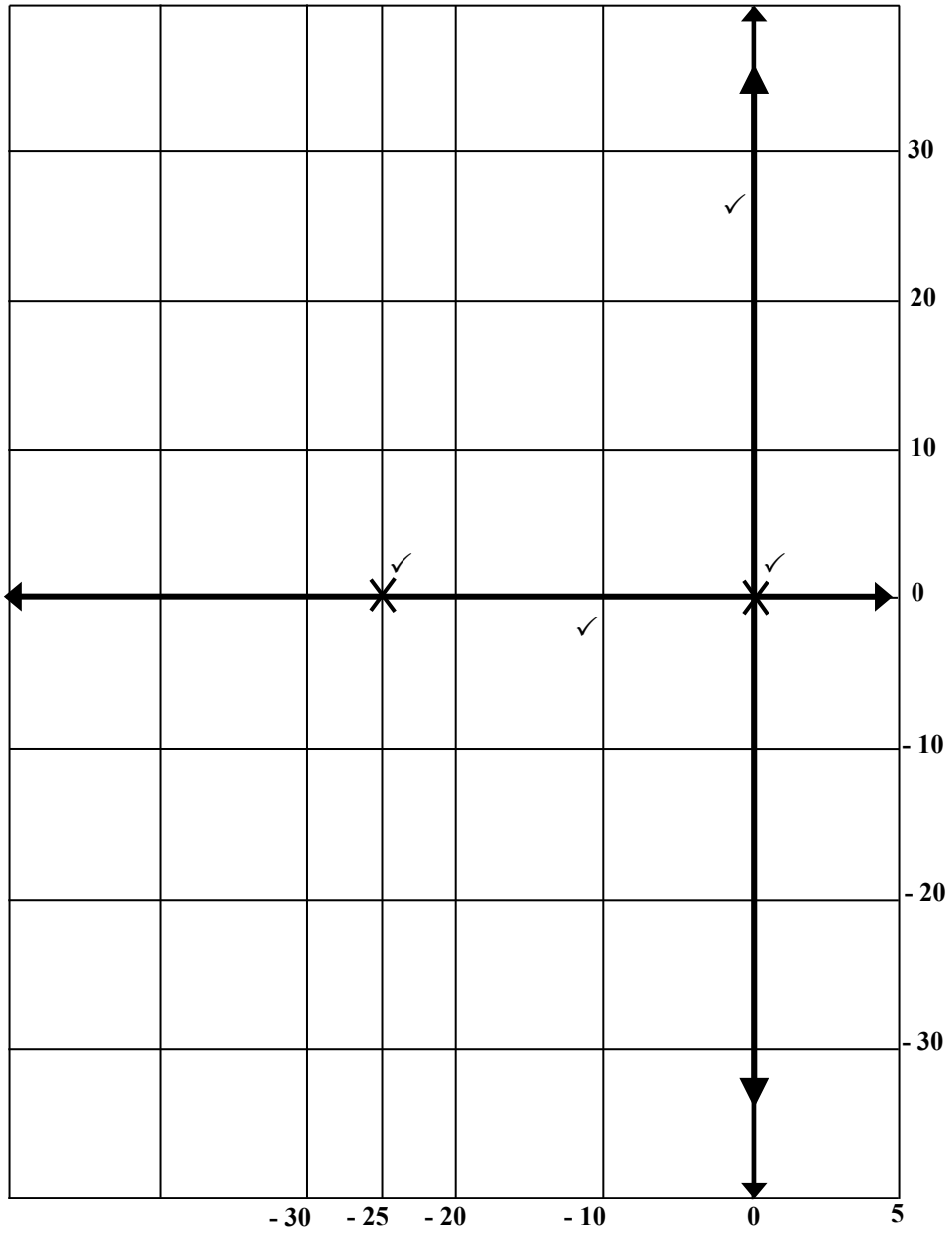
5.3
$$= \frac{(2K+1)180}{N_p - N_z}$$

$$= \frac{(2K+1)180}{2} \quad if\ K = 0\ ,\ 1,\ 2,\ 3, \infty$$

$$\psi = 90° \ when\ K = 0$$

$$\psi = 270° \ when\ K = 1 \checkmark$$ (2)

5.4



(4)
[10]

QUESTION 6

6.1 6.1.1

$$F(s)=\frac{25}{(3s+1)(2s+1)}$$

$$F(t)=\frac{A}{\tau_1-\tau_2}\quad\left[e^{\frac{-t}{\tau_1}-\frac{-t}{\tau_2}}\right]$$

$$F(t)=\frac{25}{3-2}\quad\left[e^{\frac{-t}{3}-\frac{-t}{2}}\right]\checkmark$$

$$F(t)=\frac{25}{1}\quad\left[e^{\frac{-t}{3}-\frac{-t}{2}}\right]\checkmark$$

$$F(t)=25\quad\left[e^{\frac{-t}{3}-\frac{-t}{2}}\right]\checkmark$$ (3)

6.1.2

$$F(s)=\frac{100\omega}{(s^2+\omega^2)(s+20)}$$

$$F(t)=\frac{A\omega e^{-at}}{(\omega^2+a^2)}+\frac{A}{\sqrt{\omega^2+a^2}}\sin(\omega t-\Psi)$$

$$F(t)=\frac{100\omega e^{-20t}}{(\omega^2+20^2)}+\frac{100}{\sqrt{\omega^2+20^2}}\sin(\omega t-\Psi)\checkmark$$

$$F(t)=\frac{100\omega e^{-20t}}{(\omega^2+400)}+\frac{100}{\sqrt{\omega^2+400}}\sin(\omega t-\Psi)\checkmark$$

Where

$$\Psi=\tan^{-1}\frac{\omega}{20}\quad(0<\Psi<\pi)\checkmark$$ (3)

6.2 6.2.1

$$F(t)=15te^{-2t}$$

$$F(s)=\frac{A}{(s+a)^2}$$

$$F(s)=\frac{15}{(s+2)^2}\ \checkmark\checkmark$$ (2)

6.2.2

$$F(t)=90(1-e^{\frac{t}{2}})$$

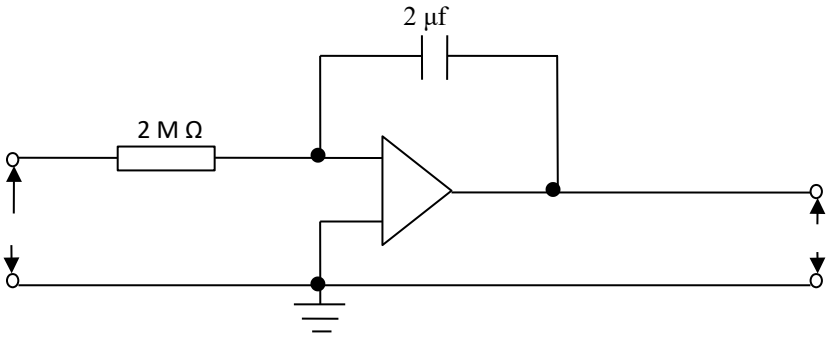
$$\frac{A}{s(\tau s+1)}$$

$$F(s)=\frac{90}{s(2s+1)}$$ ✓✓ (2)

[10]

QUESTION 7

7.1



(3)

7.2 7.2.1

$$p=\frac{1}{f}$$

$$p=\frac{1}{5\,000}$$ ✓

$$p=200\ \ \mu\ \ sec$$ ✓ (2)

7.2.2

$$T=R\times C$$
$$T=2\times10^{6}\times2\times10^{-6}$$ ✓
$$T=4\ \ sec$$ ✓ (2)

7.2.3

$$V_o(t)=-\frac{1}{RC}.\int_0^t V_i(t).dt+V_c(o)$$

$$V_o(t)=-\frac{1}{4}.\int_0^{100\times10^{-6}} 6(t).dt+0$$ ✓

$$V_o(t)=-0{,}25\left[6t\right]_0^{100\times10^{-6}}$$ ✓

$$V_o(t)=-0{,}25\left[6\times100\times10^{-6}-(6\times0)\right]$$

$$V_o(t)=-1{,}5\ \ \mu V$$ ✓ (3)

[10]

QUESTION 8

8.1
- The surface of one plate
- The dielectric constant of the medium between the plates (2)

8.2 8.2.1
A – Output
B – Insulated cover
C – Mutually-insulated metal tubes
D – Liquid
E – Tank (5)

8.2.2
- The device is static and the dielectric constant of air is 1 and that of liquid is higher, about 1,5 to 3.
- An increase in the liquid level causes an increase in the capacity of the coaxial capacitor.
- This allows for the quantity of liquid in the tank to be controlled by changing the output as the liquid level changes. (3)

[10]

QUESTION 9

9.1 Hydraulics a the power transmitted and controlled through the use of pressurised liquid or fluid such as oil, brake fluid, water, etc. (2)

9.2
- The efficiency of the liquid is affected by high temperatures
- Fluid must be kept clean and free from gas
- Danger of leaking fluids under high pressures (3)

9.3
- It is similar to a pump but is modified to ensure that full pressure can be applied to both ports.
- It is well-lubricated to allow starting under full pressure. (2)

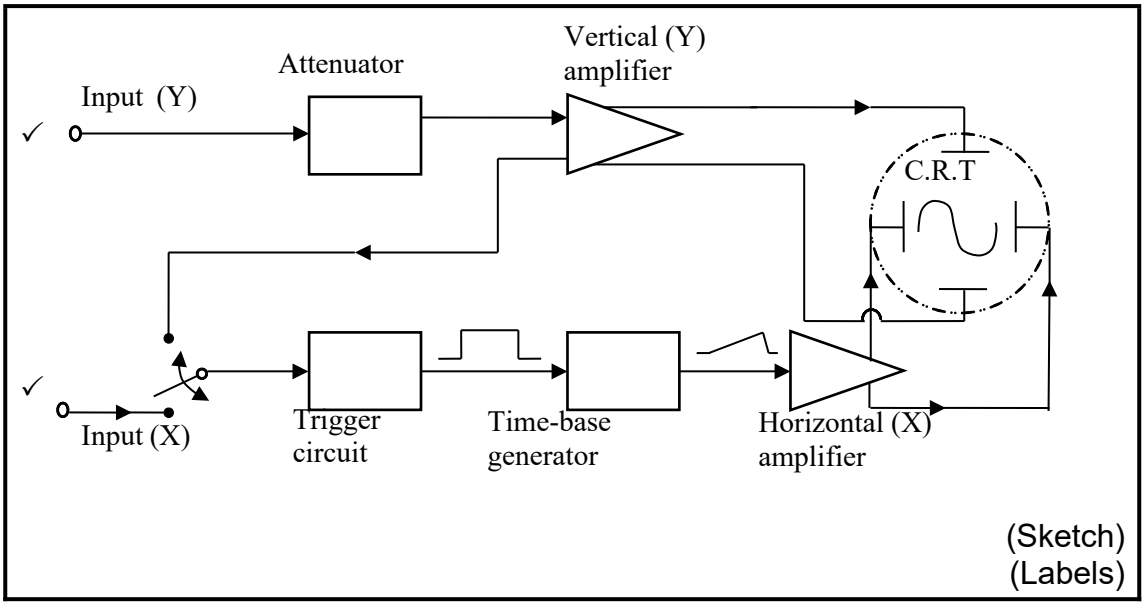
9.4
- Tank
- Pump
- Relief valve
- Flow-control valve
- Cylinder/actuator
- Filter
- Pipes (Any 3 × 1) (3)

[10]

QUESTION 10

10.1 It is a sawtooth wave generated by charging and discharging a capacitor utilising proper RC time constants. (2)

10.2



(Sketch) (3)
(Labels) (5)
[10]

TOTAL: 100

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